



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME

CLASS

CHEMISTRY

Preliminary Examination
Paper 3 Free Response

9647/03

26 Aug 2015
2 hours

Additional Materials: Data Booklet
Cover Page
Writing Papers
Annex A

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.
A Data Booklet is provided.
The use of an approved scientific calculator is expected, where appropriate.
You are reminded of the need for good English and clear presentation in your answers

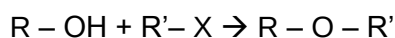
The number of marks is given in the brackets [] at the end of each question or part question.

Start each question on a fresh piece of paper.

At the end of the examination, fasten all your work securely together with the cover page provided and ensure your script is arranged in an orderly manner.

2015 Preliminary Examination Suggested Mark Scheme for Paper 3

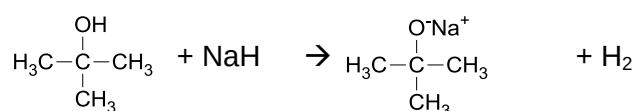
- 1 (a) The Williamson ether synthesis is an organic reaction, forming ether from a halogenoalkane and an alcohol.



This reaction is important in the history of organic chemistry because it helped in proving the structure of ethers. The alkoxide needed in the Williamson reaction is prepared by the reaction of an alcohol and sodium hydride.

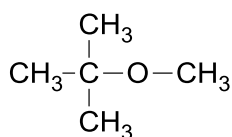
- (i) Construct a balanced chemical equation showing how this alkoxide ion (RO^-) is formed between 2-methylpropan-2-ol with a strong base sodium hydride, NaH .

[1]



[Note: the alcohol is the acid in this reaction, since it is mentioned that the NaH is the base, it will accept a proton to form H_2 gas rather than water.]

- (ii) A *tert*-butyl methyl ether can be prepared by reacting the alkoxide ion with iodomethane.



tert-butyl methyl ether

The experimental results obtained from one experiment using 0.15 mol dm^{-3} of alkoxide ion from (a)(i) and iodomethane is shown in **Annex A**.

The following information shows the lab technician's recording when the iodomethane was reacted with 0.10 mol dm^{-3} of alkoxide ion in a **separate** experiment.

Time / min	[iodomethane] / mol dm^{-3} (With $[\text{RO}^-] = 0.10 \text{ mol dm}^{-3}$)
0	0.0100
40	0.0079
80	0.0062
120	0.0049
160	0.0038
200	0.0030
240	0.0024

Using the graph in **Annex A** and the information above, determine the order of reaction with respect to

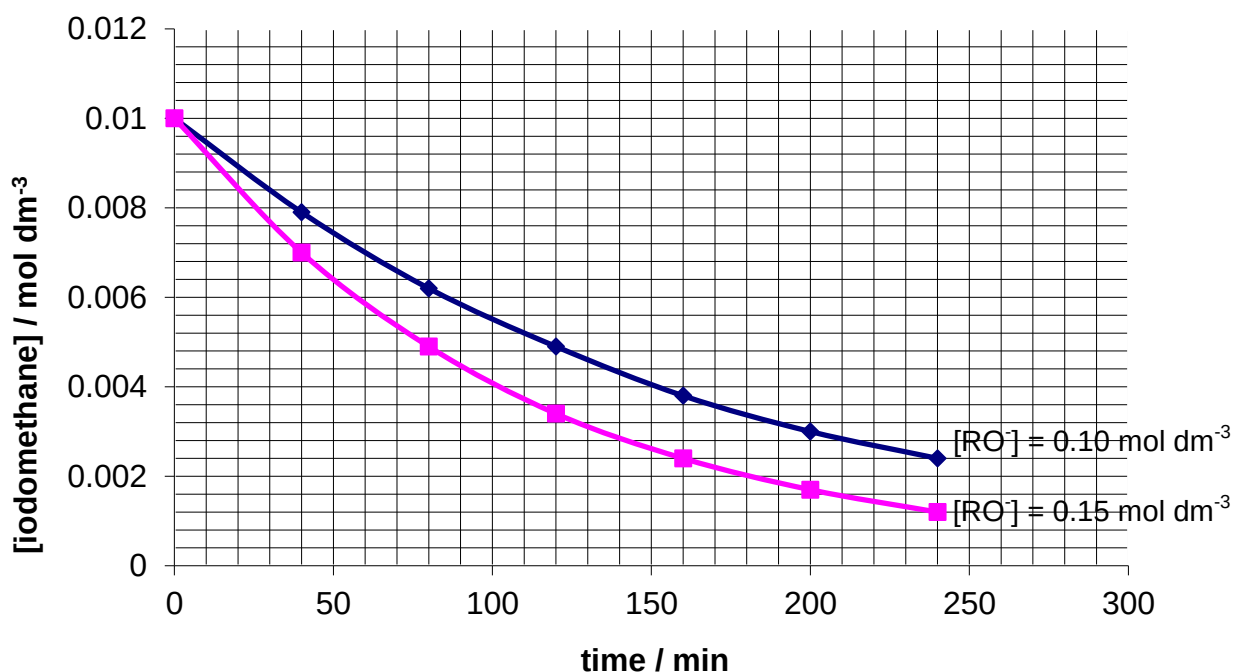
- 1 iodomethane and
- 2 alkoxide ion.

Justify your answer in each case.

(Candidate to attach **Annex A** with the answer script for submission.)

[5]

Graph of [iodomethane] / mol dm⁻³ vs time / min



(ii)

To find order of reaction wrt iodomethane

From the graph, when $[\text{OH}^-] = 0.15 \text{ mol dm}^{-3}$,
half-life is constant at 75 mins.

Hence, the order of reaction w.r.t iodomethane is 1

To find order of reaction wrt RO⁻

Let g_1 and g_2 be the gradients of the graph at $t = 0$ when $[\text{RO}^-] = 0.10 \text{ mol dm}^{-3}$ and 0.15 mol dm^{-3} respectively.

From the graphs, Initial rate for $[\text{RO}^-] = 0.10 \text{ mol dm}^{-3}$

$$\begin{aligned} \text{initial rate} &= -(g_1) \\ &= -(0.0079 - 0.0100) / (40 - 0) = \underline{5.25 \times 10^{-5} \text{ mol dm}^{-3} \text{ min}^{-1}} \end{aligned}$$

From the graphs, Initial rate for $[\text{RO}^-] = 0.15 \text{ mol dm}^{-3}$

$$\begin{aligned} \text{initial rate}_2 &= -(g_2) \\ &= -(0.0068 - 0.0100) / (40 - 0) = \underline{8.00 \times 10^{-5} \text{ mol dm}^{-3} \text{ min}^{-1}} \end{aligned}$$

Thus, when $[\text{RO}^-]$ is increased 1.5 times (from 0.10 mol dm^{-3} to 0.15 mol dm^{-3}) while keeping [iodomethane] constant, the rate is increased

$$\frac{8.00 \times 10^{-5}}{5.25 \times 10^{-5}} = \underline{1.5 \text{ times}}$$

Hence, order of reaction with respect to $\text{RO}^- = \underline{1}$

- (iii) Determine the rate constant, with units, for this reaction.
(You are to leave your answer to 2 decimal places)

[2]

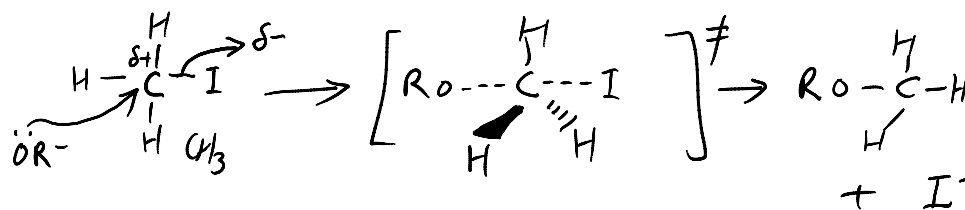
$$\text{Rate} = k [\text{iodomethane}] [\text{RO}^-]$$

$$5.25 \times 10^{-5} = k \times 0.010 \times 0.10$$

$$k = 0.0525 \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1} = \underline{\underline{0.05 \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}}}$$

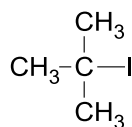
- (iv) Using the rate equation you have deduced in (a)(iii), suggest the mechanism of the reaction between iodomethane and the alkoxide ion.
(You are to include curly arrows showing the movement of the electrons, and all charges)

[2]



- (v) Suggest why the *tert*-butyl methyl ether cannot be prepared by using methoxide ion and 2-iodo-2-methylpropane.

[1]



Steric Factor: There are presence of three $-\text{CH}_3$ group which prevent the methoxide ion (which is the nucleophile) from attacking the partial positive carbon.

(Note: Williamson ether synthesis is an $\text{S}_{\text{N}}2$ pathway, reference: McMurray Organic, Chemistry Chapter 18.3)

- (b) (i) Compounds **A**, **B** and **C** share the same molecular formula $C_8H_{10}O$.

When reacted with aqueous bromine, only **A** decolorises the reddish brown solution forming a symmetrical product with formula C_8H_9BrO .

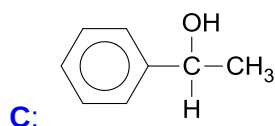
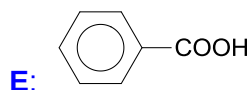
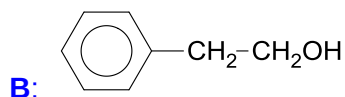
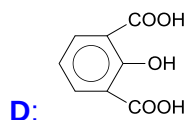
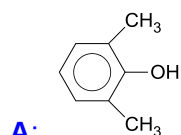
These three compounds react differently with common oxidising agents.

When oxidised with hot acidified potassium manganate(VII), **A** forms **D** which further reacts with two moles of thionyl chloride. When **B** and **C** are oxidised with the same reagent, **E** is formed. **E** is known to react with only one mole of thionyl chloride.

B and **C** forms a green solution with hot acidified sodium dichromate(VI). The product from the oxidation of **B** fizzes when sodium carbonate is added while the product of **C** gives an orange precipitate when 2,4-dinitrophenylhydrazine is added. **A**, however, is inert to acidified sodium dichromate(VI).

- (i) Deduce, with reasoning, the structural formula of compounds **A**, **B**, **C**, **D** and **E**.

[7]



A, **B** and **C** has comparable number of carbon and hydrogen and number of carbon is greater than 6 → Benzene ring is present.

A contains **phenol** as it undergoes **electrophilic substitution** with Br_2 (aq).

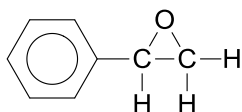
A must have **two side chains** as its oxidised product undergoes **nucleophilic substitution** with 2 mol of $SOCl_2$.

B undergoes side-chain oxidation to form **carboxylic acid** which undergo **neutralisation** with sodium carbonate. **B** contains **primary alcohol**.

C contains secondary alcohol, it undergoes side-chain oxidation to produce a **ketone** which undergoes **condensation** with 2,4-DNPH.

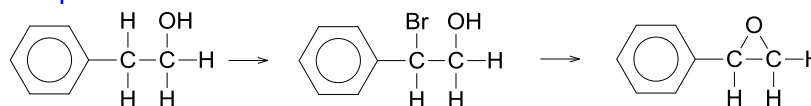
- (ii) **B** and **C** can be converted to **F**, C_8H_8O via a **two**-step reaction scheme which involves the Williamson ether synthesis. The yield of **F** is known to be low. Draw the displayed formula of compound **F**.

[1]



For your information:

The two step reaction scheme using compound **B** as the starting compound.



F is an epoxide, an epoxide is a cyclic ether with a three-atom ring. This ring approximates an equilateral triangle which makes it strained and highly reactive.

- (iii) Suggest why the yield of compound **F** is low.

[1]

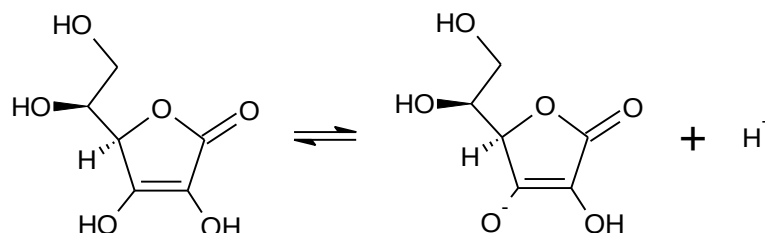
This ring approximates an equilateral triangle which makes it strained and highly reactive/unstable.

Or

To form epoxide **F**, it needs to undergo free radical substitution which generates a mixture of products resulting in lower yield.

[Total: 20]

- 2 Vitamin C, $C_6H_8O_6$, is also known as ascorbic acid. The first acid dissociation is as follows (pK_a value 4.10):



- (a) (i) With reference to its structure, explain why vitamin C has a pK_a value of 4.10.

The ascorbate anion is resonance stabilised by the delocalisation of the negative charge over the neighbouring C atoms and the C=O in the ascorbate anion, hence stabilising the ascorbate anion relative to the molecule. This causes a greater tendency for Vitamin C to ionise.

- (ii) A 500mg tablet of Vitamin C is swallowed and dissolved in the stomach, which may be assumed to contain 1.0 dm^3 of 0.10 mol dm^{-3} hydrochloric acid.

Calculate the percentage of Vitamin C that is ionised in the stomach under these conditions.

$$[C_6H_8O_6 \text{ in } 500 \text{ mg}] = \frac{500 \times 10^{-3}}{176} = 2.84 \times 10^{-3} \text{ mol dm}^{-3}$$

	$C_6H_8O_6$	$\square$	$C_6H_7O_6^-$	+	H^+
Initial [] / mol dm^{-3}	2.84×10^{-3}		0		0.1
Change in [] / mol dm^{-3}	$-x$		$+x$		$+x$
Final [] / mol dm^{-3}	$2.84 \times 10^{-3} - x$		x		$0.1 + x$

Assuming $0.1 \gg x$

$$(0.1)(x) / (2.84 \times 10^{-3} - x) = 7.94 \times 10^{-5}$$

$$x = 2.26 \times 10^{-6} \text{ mol dm}^{-3}$$

$$\begin{aligned} \% \text{ ionised} &= (2.26 \times 10^{-6} / 2.84 \times 10^{-3}) \times 100\% \\ &= \underline{\underline{0.0796\%}} \end{aligned}$$

- (iii) Explain qualitatively how the extent of the ionisation of Vitamin C changes as it moves from the stomach into the bloodstream.

[6]

As vitamin C moves from the stomach into the bloodstream, there is a decrease in $[H^+]$.

By Le Chatelier's Principle, the position of equilibrium will shift to increase the concentration of Vitamin C.

The forward reaction is favoured and extent of ionisation of vitamin C will be increased until a new equilibrium is reached.

- (b) (i) 500 mg of the ascorbic acid was dissolved and made up to 250 cm³ in a standard flask. 25.0 cm³ of the acid was then pipetted and titrated with x mol dm⁻³ of sodium hydroxide. 16.40 cm³ of base was required for complete neutralisation.

Using the information provided, determine the initial pH of the acid and the concentration of the sodium hydroxide, in g dm⁻³ that is required for the complete neutralisation.

$$\text{Amt of C}_6\text{H}_8\text{O}_6 \text{ in 500 mg} = \frac{500 \times 10^{-3}}{176} = 2.84 \times 10^{-3} \text{ mol}$$

$$[\text{H}^+] = \sqrt{K_a \times [\text{acid}]} = \sqrt{10^{-4.10} \times \frac{2.84 \times 10^{-3}}{\frac{250}{1000}}} = 9.499 \times 10^{-4} \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg(9.499 \times 10^{-4}) = 3.02$$

Let C₆H₈O₆, HA



$$\text{Amt of HA in 25.0 cm}^3 = 2.84 \times 10^{-4} \text{ mol}$$

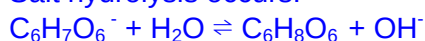
$$\text{Amt of NaOH required} = 2.84 \times 10^{-4} \text{ mol}$$

$$[\text{NaOH}] = \frac{2.84 \times 10^{-4}}{\frac{16.40}{1000}} = 0.01731 \text{ mol dm}^{-3}$$

$$[\text{NaOH}] \text{ in g dm}^{-3} = 0.01731 \times (23.1 + 16 + 1) = 0.694 \text{ g dm}^{-3}$$

- (ii) With an aid of chemical equation explain why the end point pH is basic in nature.

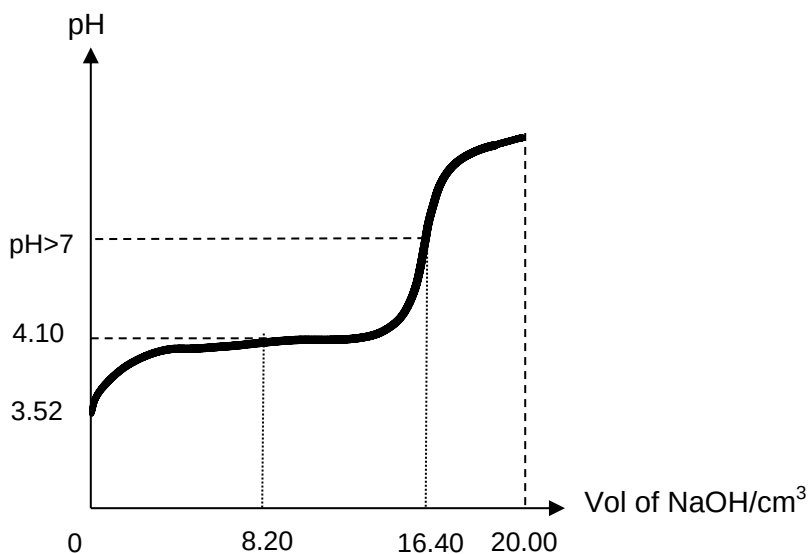
Salt hydrolysis occurs.



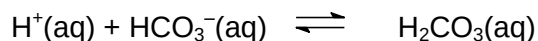
Since [OH⁻] > [H⁺], end point pH is basic

- (iii) Using the information from (a), and (b)(i), sketch the pH-volume graph you will expect to obtain when 20 cm³ of sodium hydroxide is added to 25.0 cm³ of ascorbic acid . Label the various key points on the curve.

[7]

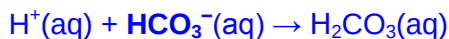


- (c) The following equilibrium reaction acts as buffer in blood.



Explain, with equations, clearly how this reaction minimises the change in pH on the addition of acid or alkali

When **small amount of acid** is added, H⁺ reacts with the **hydrogencarbonate ion** to form **carbonic acid**.



When **small amount of base** is added, OH⁻ reacts with the **carbonic acid** to form **hydrogencarbonate ion**.



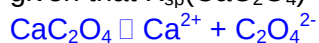
[2]

- (d) It was reported in recent studies that there may be a link between excessive

dosage of Vitamin C and kidney stone formation.

The main composition of kidney stone is calcium oxalate, CaC_2O_4 , whereby the oxalate ion is produced during the metabolism of Vitamin C in the body.

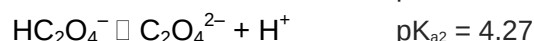
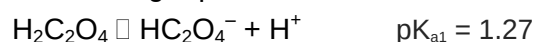
- (i) Calculate the solubility, in mol dm^{-3} , of calcium oxalate in pure water given that $K_{\text{sp}}(\text{CaC}_2\text{O}_4) = 6.7 \times 10^{-4} \text{ mol}^2 \text{ dm}^{-6}$



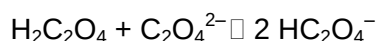
$$K_{\text{sp}} = s^2$$

$$s = \underline{0.0259 \text{ mol dm}^{-3}}$$

- (ii) By considering the following equilibria



Derive an expression, and calculate a value for, the equilibrium constant K_c for



[3]

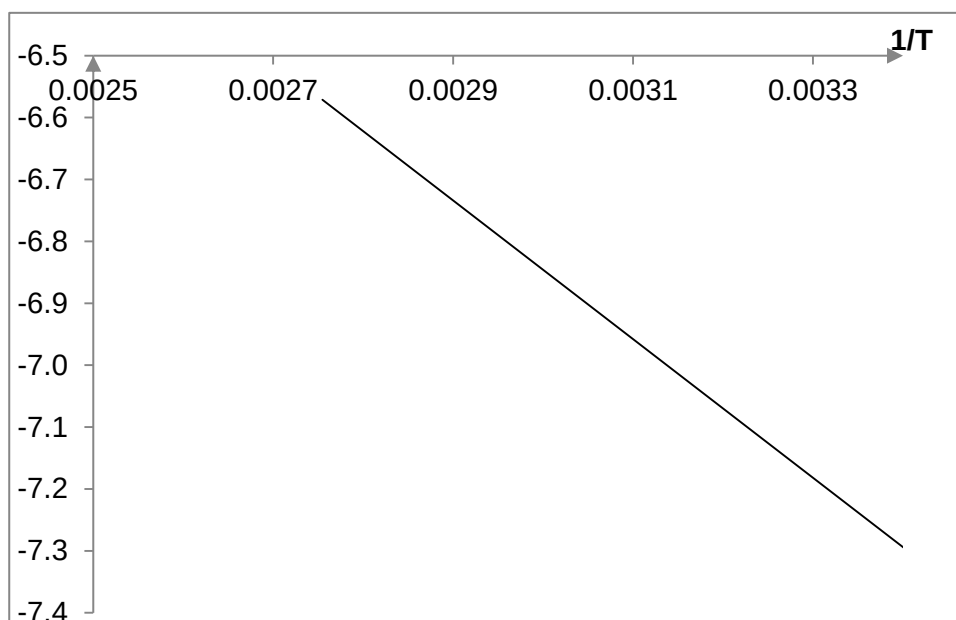
$$K_c = \frac{[\text{HC}_2\text{O}_4^-]^2}{[\text{H}_2\text{C}_2\text{O}_4][\text{C}_2\text{O}_4^{2-}]}$$

$$= 10^{-1.27} / 10^{-4.27}$$

$$= \underline{1000}$$

- (e) A plot of $\ln s$ against $1/T$ can be obtained to investigate the effects of the solubility (s) of calcium oxalate with varying temperatures (in Kelvin). With reference to the graphical data below, discuss how an increase in temperature may affect the solubility of calcium oxalate.

[2]



From the graph, as temperature of the reaction increases, solubility increases.

- average kinetic energy of the solvent molecules increases [1]
- solvent molecules are able to overcome the stronger electrostatic forces of attraction between the Ca^{2+} and $\text{C}_2\text{O}_4^{2-}$ ions.
- solubility increases. [1]

[Total : 20]

- 3 (a) Calcium carbonate is used heavily in the construction industry as a building material. The raw form limestone has to be purified through a series of processes. The first step involves the decomposition of calcium carbonate known as calcining which happens in large chambers called kilns.

The kiln is made of steel, lined with refractory bricks. Calcium carbonate is fed in from the top. Fuel for heating is fed through the sides of the kiln through tubes called lances.

The kiln consists of three principle zones as shown in the diagram:

Preheating zone	CaCO_3 is preheated to 1200 K.
Burning zone	CaCO_3 decomposes to form calcium oxide and carbon dioxide at 1500 K.
Cooling zone	Heat in the CaO leaving the burning zone is used to preheat the air required for combustion to 600-750 K.

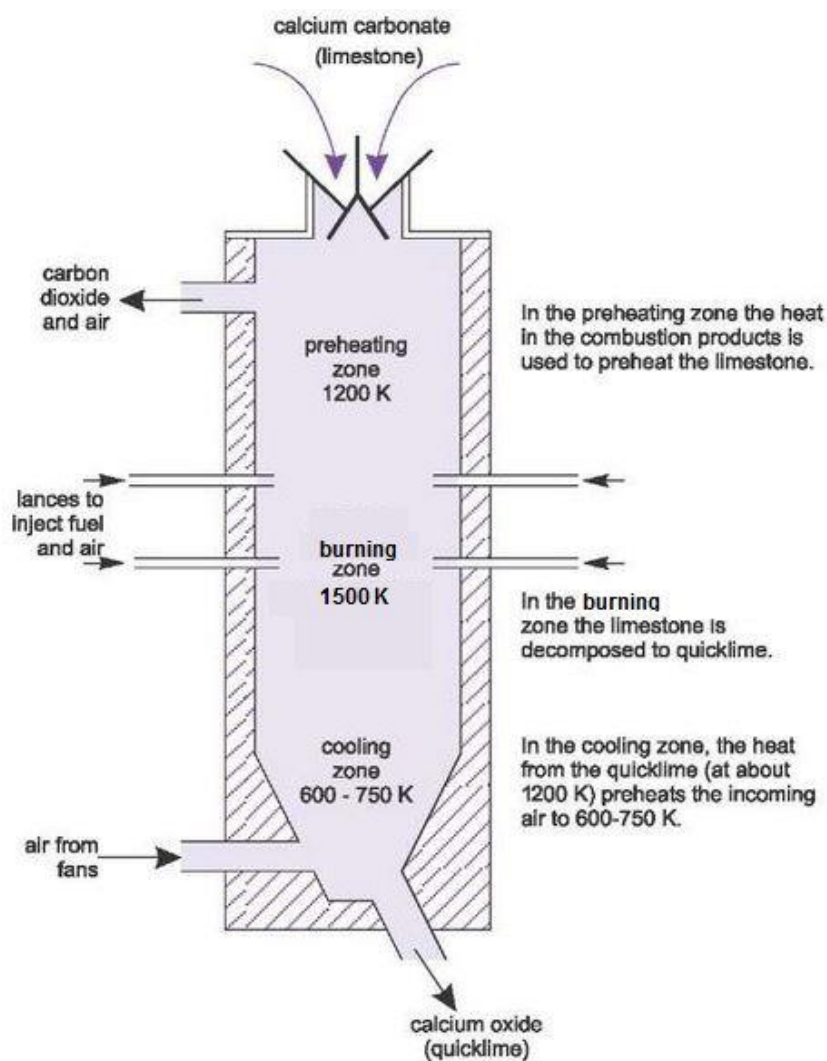


Figure 1: Structure of kiln

- (i) Suggest a reason for pumping more air into the burning zone.

This is to provide enough air/oxygen for complete combustion of the fuel.

- (ii) Why is air from the cooling zone used to preheat the incoming air?

This is to maximize burning efficiency / reduce the amount of fuel used to reduce cost of production.

[2]

- (b) The thermal decomposition temperature is the temperature when CaCO_3 starts to decompose and eventually reaches equilibrium. To determine this temperature, a piece of CaCO_3 was heated strongly to allow decomposition to occur and subsequently allowed to cool to room temperature. The piece was then placed in 25 cm^3 of excess dilute hydrochloric acid. The temperature of the solution was recorded in Table 1.

Table 1

Mass of boiling tube and CaCO_3 / g	21.591
Mass of empty boiling tube / g	20.402
Initial temperature of solution / $^{\circ}\text{C}$	28.3
Final temperature of solution / $^{\circ}\text{C}$	47.1

Given :

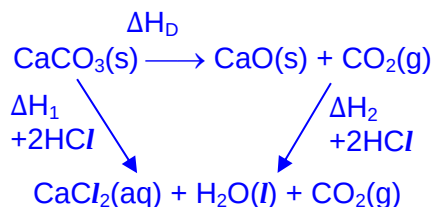
- The reaction between calcium carbonate and acid is 85% efficient.
- $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g}) \quad \Delta H_1 = -14.8 \text{ kJ mol}^{-1}$

- (i) Using the information given above and with a suitable energy cycle, calculate the enthalpy change of decomposition of calcium carbonate, leaving your answer to 1 decimal place in kJ mol^{-1} .



$$Q = 25 \times 4.18 \times 18.8 \times (100/85) = 2.311 \text{ kJ}$$

$$\Delta H_2 = \frac{-Q}{\left(\frac{1.189}{100.1}\right)} = \frac{-2.311}{\left(\frac{1.189}{100.1}\right)} = -194.6 \text{ kJ mol}^{-1}$$



$$\Delta H_D = -14.8 - (-194.6) = \underline{\underline{+179.8 \text{ kJ mol}^{-1}}}$$

- (ii) Given that ΔS is $+161 \text{ J mol}^{-1} \text{ K}^{-1}$ for the decomposition of calcium carbonate and using your answer in (b)(i), calculate the thermal decomposition temperature, in K.

At equilibrium, $\Delta G = 0$

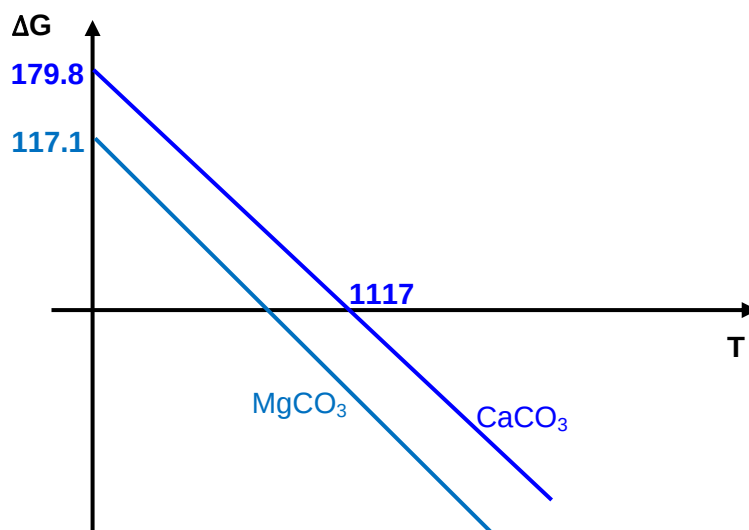
$$0 = 179.8 - T(161/1000)$$

$$T = 1117 \text{ K}$$

- (iii) Suggest why a higher temperature is used in the burning zone for the calcination of CaCO_3 .

The decomposition of CaCO_3 is endothermic. An increase in temperature would cause the position of equilibrium to shift right to absorb heat, favouring the endothermic reaction and increasing the yield of CaO .

- (iv) Hence, sketch the graph of ΔG against T , labelling clearly the thermal decomposition temperature.



- (v) Another material used heavily in construction is MgCO_3 . Crude MgCO_3 also undergoes calcination to produce MgO .

Given that $\Delta H_D(\text{MgCO}_3) = +117.1 \text{ kJmol}^{-1}$, sketch on the **same** graph the graph of ΔG against T for the calcination of MgCO_3 . Explain your reasoning and include any assumptions made.

Assumption: ΔS is the same for both reactions as no of moles of gaseous products produced is the same.

(On graph above)

(concept: thermal stability of Gp II carbonates)

Down the group,

Cationic charge remains constant. while Cationic radius of Ca^{2+} is larger than Mg^{2+}

Charge density, and hence polarising power: $\text{Mg}^{2+} > \text{Ca}^{2+}$
Polarising effect on the anion: $\text{Mg}^{2+} > \text{Ca}^{2+}$

Distortion of anion electron cloud: $\text{Mg}^{2+} > \text{Ca}^{2+}$

Less energy is required to overcome the C–O bond for MgCO_3

A lower temperature is required to decompose MgCO_3 .

[12]

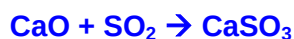
- (c) The product of calcination, calcium oxide, is important in agriculture to create the optimum soil conditions for crop growth. Due to its corrosive nature, calcium oxide is usually not added directly to soil. Instead, it is mixed with water in a storage tank to form calcium hydroxide before adding to soil. In an unfortunate accident, calcium hydroxide leaks from a storage tank to a nearby plot. The pH of the soil was found to be 12.4. Calculate the K_{sp} of calcium hydroxide.

$$\begin{aligned} [\text{OH}^-] &= 10^{-(14 - 12.4)} = 2.51 \times 10^{-2} \text{ mol dm}^{-3} \\ [\text{Ca}^{2+}] &= 0.5 \times 2.51 \times 10^{-2} = 1.26 \times 10^{-2} \text{ mol dm}^{-3} \\ K_{sp} &= [\text{Ca}^{2+}][\text{OH}^-]^2 \\ &= (1.26 \times 10^{-2})(2.51 \times 10^{-2})^2 \\ &= 7.92 \times 10^{-6} \text{ mol}^3 \text{ dm}^{-9} \end{aligned}$$

[3]

- (d) Another application of calcium oxide is also used in large quantities to clean up the environment. One example is the removal of flue gas (sulfur dioxide) from the exhaust of coal. Before being released into the atmosphere, exhaust gas from the burning of coal is passed through a spray of very finely grounded CaO suspended in water to produce calcium sulfite.

- (i) Write an equation for the reaction between calcium oxide and flue gas.



- (ii) Name the type of reaction for (d)(i).

Neutralisation

- (iii) Why is it important to remove sulfur dioxide from the exhaust gas of coal?

Sulfur dioxide can dissolve in rain water to form acid rain.

[3]

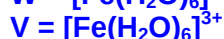
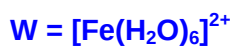
[Total : 20 m]

- 4 Iron is a metal in the first transition series. It is, by mass, the most common element on Earth, forming much of Earth's outer and inner core. Like most transition elements, iron and its compounds exist in an array of colours. Also, iron is widely used in industries because of its diversity of chemical reactions. It is even found in human body as haemoglobin or more commonly known as red blood cells.

- (a) Two separate experiments were carried out to explore the chemical reactions of iron and its compounds.

Iron metal reacts with reagent **X** to give a pale green solution **W**, which on reaction with reagent **Y** is converted to a pale yellow solution **V**. When **W** is boiled with an excess of sodium cyanide, a yellow solution **U** is produced, from which pale yellow crystals can be obtained by evaporation. When solution **V** reacts with sodium carbonate, effervescence and a red-brown solid is seen. A blood-red solution is also obtained when a few drops of reagent **Z** is added to solution **V**.

- (i) State the formulae of the cations in **W** and **V** and suggest identities for reagents **Y** and **Z**.



Y: KMnO_4 or Cl_2 or H_2O_2 (or in words)

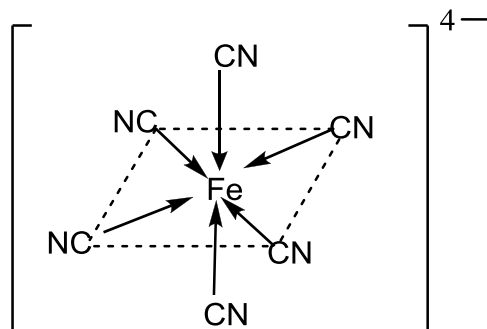
Z: aqueous KSCN (or NH_4SCN)

- (ii) Describe briefly the chemistry for the observations made when solution **V** reacts with sodium carbonate.

(Complete) hydrolysis occurs as the high charge density of Fe^{3+} polarises the surrounding water molecule, weakens the O-H bond in water, liberating H^+ into the solution causing the solution to be acidic.

Subsequently, the acidic H^+ is neutralised by the weak base, sodium carbonate to liberate CO_2 gas and form $\text{Fe}(\text{OH})_3(\text{H}_2\text{O})_3$.

- (iii) Draw the structure of the iron-containing species in **U**, showing clearly the disposition of the atoms.



- (b) The use of the Data Booklet and the writing of balanced chemical equations are required in this question.

In a separate experiment, when solution **V** is added to aqueous potassium iodide, a brown solution **T** is obtained. Solution **T** is found to catalyse the reaction between iodide and peroxodisulfate, $\text{S}_2\text{O}_8^{2-}$, whereas solution **U** has no impact on the rate of this reaction. When aqueous sodium thiosulfate is added to solution **T** followed by small amounts of ammonia, the brown colour decolourises and a green precipitate, **S**, is observed. On standing, the green precipitate turns red-brown, **R**.

- (i) By considering the species present in solution **T**, explain how it is capable of acting as a catalyst and describe its mechanism.

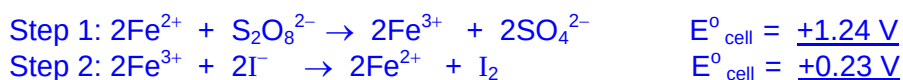
When solution **V** containing Fe^{3+} is added to I^- , I^- is oxidised to brown iodine by Fe^{3+} while Fe^{3+} is reduced to Fe^{2+} . Hence, solution **T** contains iodine and Fe^{2+} .

This redox reaction is feasible as $E^\circ_{\text{cell}} = 0.77 - 0.54 = +0.23\text{V} > 0$
 $2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$

Solution **T** containing Fe^{2+} can act as a **homogenous catalyst** for the reaction as Fe^{2+} can exist in **variable oxidation states** thus facilitating the formation of reaction intermediates via alternative pathways of lower E_a .

The **activation energy** for the reaction between iodide and peroxodisulphate, $\text{S}_2\text{O}_8^{2-}$ is **lowered** as reactions are now between **oppositely charged ions** and are not between two negatively charged ions.

The catalysed pathway involves 2 steps with each step having a lower activation energy:



- (ii) Suggest why solution **U** has no impact on the rate of reaction.

Solution **U** contains complex ion, $[\text{Fe}(\text{CN})_6]^{4-}$, is formed between Fe^{2+} and excess CN^- . **Ligand exchange** occurs to form a **more stable complex**.



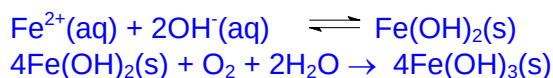
However, solution **U** contains complex ion, $[\text{Fe}(\text{CN})_6]^{4-}$. [The redox potential of $E^\circ([\text{Fe}(\text{CN})_6]^{3-}/[\text{Fe}(\text{CN})_6]^{4-})$ is $+0.36\text{V}$, which is **not within the required range**. Furthermore, both $[\text{Fe}(\text{CN})_6]^{4-}$ and I^- are **negatively charged** and the **activation energy** of such a reaction is still **high**.

[The E° value of $\text{Fe}^{2+}/\text{Fe}^{3+}$ of $+0.77\text{V}$ lies between that of $E^\circ(\text{I}_2/\text{I}^-) = 0.54\text{V}$ and $E^\circ(\text{S}_2\text{O}_8^{2-}/\text{SO}_4^{2-}) = +2.01\text{V}$.]

- (iii) Describe the formation of the coloured precipitates **S** and **R**.

When aq. ammonia is added in small amounts (provides OH⁻) to solution **T** containing Fe²⁺ ions, a (dirty) green precipitate of Fe(OH)₂, **S**, is formed.

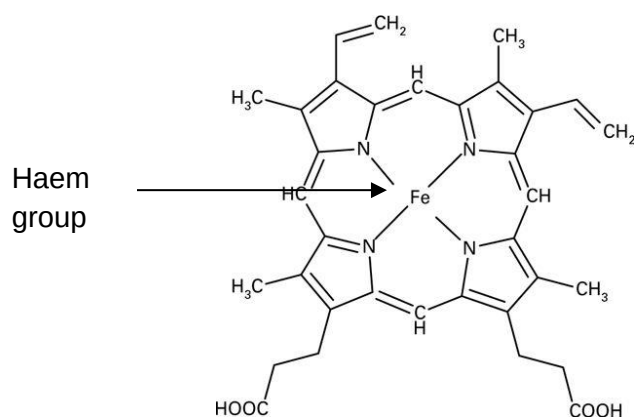
Hence Fe(OH)₂ is oxidised by air to red-brown solid, **R**, Fe(OH)₃.



This redox reaction is feasible as $E^{\circ}_{\text{cell}} = 0.40 - (-0.56) = +0.96\text{V} > 0$

[8]

- (c) Haemoglobin is the protein responsible for the transport of oxygen in humans. It is common knowledge that iron plays a critical role in the function of the protein. Haemoglobin is commonly found in red blood cells, and exhibits characteristics of both the tertiary and quaternary structures of proteins. The figure below shows the quaternary structure of haemoglobin with the haem group found in haemoglobin.



- (i) With reference to haemoglobin, describe briefly the quaternary structure of proteins.

Quaternary structure is the precise arrangement of 2 or more polypeptide chains in oligomer/highly complex proteins, held together by R group interactions. Haemoglobin consists of 2 α sub-units and 2 β sub-units.

- (ii) State two kinds of predominant attractive forces present between the iron atom and the four neighbouring nitrogen atoms.

Covalent dative / co-ordinate and Ionic bonding

- (iii) It has been found that all haemoglobin molecules in humans have identical tertiary structure. Explain why this is so.

They all contain the same sequence of amino acids (primary structure), hence the R-group interactions all occur in the same positions giving rise to the same tertiary structure.

- (iv) Do you expect haemoglobin to be soluble in water? Explain your reasoning.

Haemoglobin is soluble in water as it is a globular protein where the folding of polypeptide chains caused the hydrophobic R groups to be internally located away from the aqueous surroundings while the hydrophilic R groups are located on the surface of the protein.

- (v) Name a poison and describe how it hinders the function of haemoglobin.

Poison: Carbon monoxide / Cyanide

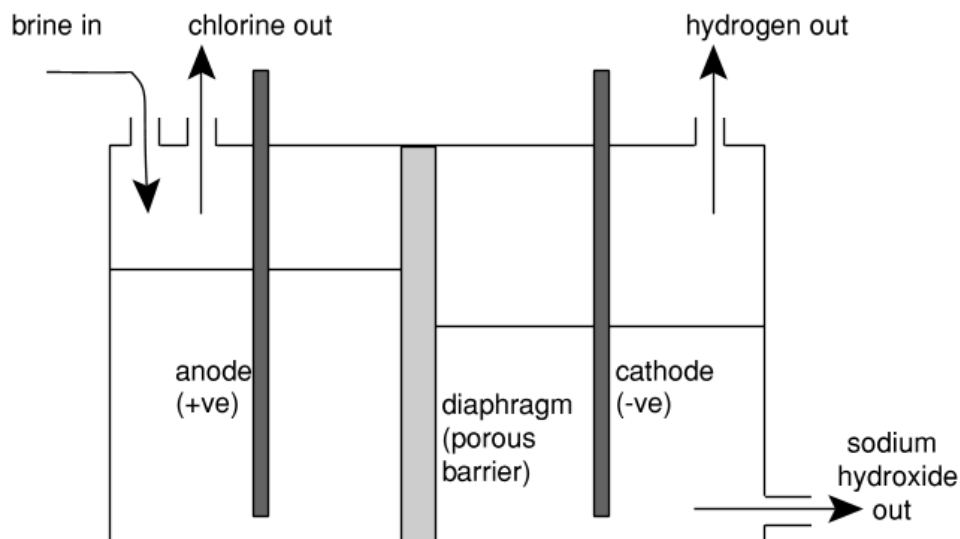
Carbon monoxide / Cyanide forms stronger dative bond with iron atom in haemoglobin more readily than O₂ molecule (since iron-CO bond is stronger than iron-O₂ bond). Also, the reaction is irreversible.

[7]

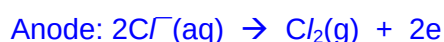
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- 5 Chlorine, which is a Group VII element, forms a variety of oxides and oxoanions. It has been extensively used in the manufacture of many compounds.

(a) Chlorine is produced commercially via electrolysis of concentrated sodium chloride, also known as brine, using inert electrodes.



- (i) Give the ion-electron equation for the reactions which occur at each of the electrodes in the electrolysis of brine.



- (ii) Calculate the mass of chlorine produced if a current of 5 A was passed through the electrolytic cell for 7 hours.

$$Q = I \times t = 5 \times 7 \times 60 \times 60 = 1.26 \times 10^5 \text{ C}$$

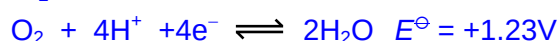
$$Q = n_e \times F$$

$$\text{Amount of electrons passed} = \frac{1.26 \times 10^5}{96500} = 1.306$$

$$\text{Amount of chlorine} = \frac{1.306}{2} = 0.6529$$

$$\text{Mass of chlorine produced} = 0.6529 \times 2 \times 35.5 = \underline{\underline{46.4 \text{ g}}}$$

- (iii) Using relevant data from the *Data Booklet*, suggest why dilute sodium chloride solution cannot be used in the production of chlorine.



When dilute sodium chloride is used, the $E^\ominus(\text{O}_2/\text{H}_2\text{O})$ is less positive than $E^\ominus(\text{Cl}_2/\text{Cl}^-)$ and H_2O is preferentially oxidised at the anode to give O_2 gas.

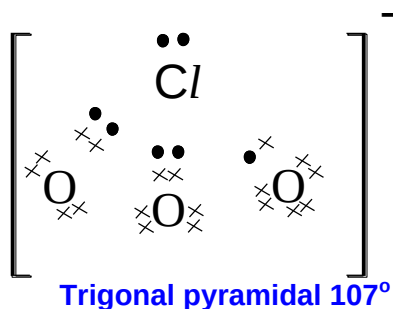
(b) If the electrolytic cell in (a) is heated, two of the products formed will react in a 1 : 2 ratio to give three products.

- (i) State the type of reaction involved and write an equation, with state symbols, for the reaction.

Disproportionation



- (ii) Draw a dot-and-cross diagram for the oxoanion formed and hence indicate the bond angle and its shape.



- (iii) A series of standard reduction potentials involving the chlorine oxoanions, in alkaline medium, are given below.

Reduction Potential half-equation	E°/V
$\text{ClO}^- + \text{H}_2\text{O} + 2\text{e}^- \rightleftharpoons \text{Cl}^- + 2\text{OH}^-$	-0.89
$\text{ClO}_2^- + \text{H}_2\text{O} + 2\text{e}^- \rightleftharpoons \text{ClO}^- + 2\text{OH}^-$	-0.67
$\text{ClO}_3^- + \text{H}_2\text{O} + 2\text{e}^- \rightleftharpoons \text{ClO}_2^- + 2\text{OH}^-$	-0.33
$\text{ClO}_4^- + \text{H}_2\text{O} + 2\text{e}^- \rightleftharpoons \text{ClO}_3^- + 2\text{OH}^-$	-0.35

Using the data given above and relevant data from the *Data Booklet*, predict the final product formed when excess zinc metal is added to an alkaline solution of the oxoanion proposed in (ii).

Reaction of zinc metal with chlorate(V):

$$\begin{aligned}
 E_{\text{cell}}^\ominus &= E^\ominus(\text{ClO}_3^-/\text{ClO}_2^-) - E^\ominus(\text{Zn}^{2+}/\text{Zn}) \\
 &= -0.33 - (-0.76) \\
 &= \mathbf{+0.43\text{ V}}
 \end{aligned}$$

Thus, Zn is able to reduce ClO_3^- to ClO_2^- .

Reaction of zinc metal with chlorate(III):

$$\begin{aligned}
 E_{\text{cell}}^\ominus &= E^\ominus(\text{ClO}_2^-/\text{ClO}^-) - E^\ominus(\text{Zn}^{2+}/\text{Zn}) \\
 &= -0.67 - (-0.76) \\
 &= \mathbf{+0.09\text{ V}}
 \end{aligned}$$

Thus, Zn is able to reduce ClO_2^- to ClO^- .

Reaction of zinc metal with chlorate(I):

$$\begin{aligned} E_{\text{cell}}^{\ominus} &= E^{\ominus}(\text{ClO}^{-}/\text{Cl}^{-}) - E^{\ominus}(\text{Zn}^{2+}/\text{Zn}) \\ &= -0.89 - (-0.76) \\ &= -0.13 \text{ V} \end{aligned}$$

Rxn between ClO^{-} and Zn^{2+} is not energetically feasible

OR Zn is not able to reduce ClO^{-} to Cl^{-} .

Hence, zinc will be oxidised to Zn^{2+} and chlorate(V) will be reduced to chlorate(I)/ ClO^{-} .

[8]

(c) The use of the Data Booklet is relevant to this question.

Describe **briefly** a simple experiment to show that chlorine is a stronger oxidising agent than iodine.

In your experiment, you are

- 1 to use a transition metal ion.
- 2 **not allowed** to use halide.
- 3 to write down all visible observations.

Detailed mention of quantities and apparatus is not required in your answer.

Add 2cm^3 of aqueous iron(II) sulphate separately to 1cm^3 of aqueous chlorine and iodine in separate test tubes.

Chlorine is strong enough an oxidising agent to oxidise Fe^{2+} to Fe^{3+} hence the pale green Fe^{2+} solution will turn into yellow Fe^{3+} .

But iodine is too weak an oxidising agent to oxidise Fe^{2+} to Fe^{3+} hence no change to the pale green Fe^{2+} solution.

(Quoting of $E^{\ominus}_{\text{cell}}$ values is acceptable in this question.)

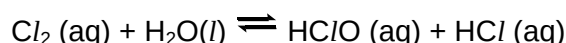
($\text{Cr}^{3+} \rightarrow \text{Cr}_2\text{O}_7^{2-}$ don't accept as the overall E value is too close to zero

($\text{Mn}^{2+} \rightarrow \text{MnO}_2$)

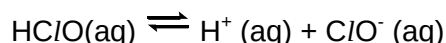
[2]

(d) Chlorine is able to take on the role of nucleophiles and electrophiles, hence are commonly used in organic reactions to synthesise new compounds.

The reaction of chlorine with water forms chloric(I) acid, HClO .



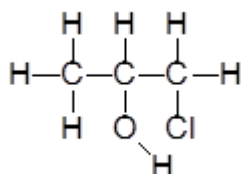
Chloric (I) acid ionises as a weak acid:



Both chloric(I) acid and hydrogen chloride react with propene. In the reaction between chloric(I) acid and propene, chlorine is added to carbon-1 of propene in the major product. However, in the reaction between hydrogen chloride and propene, chlorine is added to carbon-2 of propene in the major product.

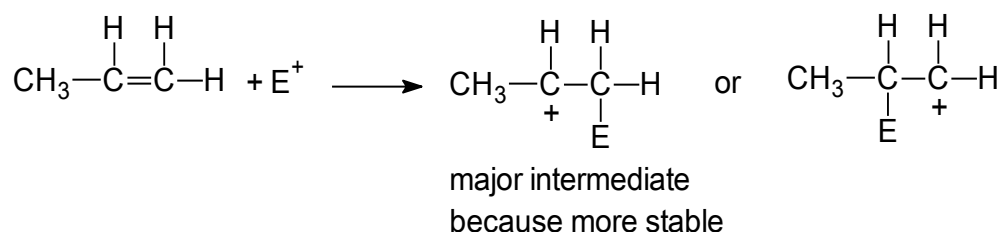
(i) Draw the displayed formula of the major product formed between chloric(I)

acid and propene.



- (ii) Explain briefly why chlorine from chloric(I) acid and chlorine from hydrogen chloride add to different carbon atoms in their reactions with propene.

Propene undergoes electrophilic addition and the first step involves the addition of the electrophile to carbon-1 as the more stable carbocation is formed:



In HOCl , Cl is δ^+ so that Cl^+ is the electrophile and Cl^+ adds to the carbon-1 to form the more stable carbocation.

Subsequent addition of OH^- results in the formation of the major product with Cl added to carbon-1.

In HCl , Cl is δ^- so that the electrophile is H^+ rather than Cl^+ . Hence H^+ adds to carbon-1 to form the more stable carbocation.

Subsequent addition of Cl^- to carbon-2 results in the formation of the major product with Cl added to carbon-2.

[4]
[Total : 20 m]

END OF SOLUTIONS